Assignment 1.3

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Task 1

```
# Fibonacci Sequence Generator (Without Functions)

# Accept user input
n = int(input("Enter number of terms: "))

# First two Fibonacci numbers
a = 0
b = 1

print("Fibonacci sequence:")

if n <= 0:
    print("Please enter a positive number.")
elif n == 1:
    print(a)
else:
    print(a)
    print(b)

    count = 2
```

```
    while count < n:
        c = a + b
        print(c)
        a = b
        b = c
        count += 1
```

Output

```
Fibonacci sequence:
0
1
1
2
3
5
8
```

Task 2

```
# Optimized Fibonacci Sequence Generator

n = int(input("Enter number of terms: "))

a, b = 0, 1

print("Fibonacci sequence:")

for i in range(n):
    print(a)
```

```
    a, b = b, a + b
```

Output

```
Fibonacci sequence:
0
1
1
2
3
5
8
```

Task 3

```
# Fibonacci Sequence Generator Using Functions

def fibonacci(n):
    """
    Generates and prints Fibonacci sequence up to n terms.
    """
    a, b = 0, 1

    for i in range(n):
        print(a)
        a, b = b, a + b


# Main program
n = int(input("Enter number of terms: "))
```

```
if n <= 0:
    print("Please enter a positive number.")
else:
    print("Fibonacci sequence:")
    fibonacci(n)
```

Task 4

```
# Procedural Fibonacci Program

n = int(input("Enter number of terms: "))

a, b = 0, 1

print("Fibonacci sequence:")

for i in range(n):
    print(a)
    a, b = b, a + b



# Modular Fibonacci Program

def fibonacci(n):
    a, b = 0, 1
    for i in range(n):
```

```
        print(a)
        a, b = b, a + b


n = int(input("Enter number of terms: "))

print("Fibonacci sequence:")
fibonacci(n)
```

Task 5

1

```
# Iterative Fibonacci Implementation

n = int(input("Enter number of terms: "))

a, b = 0, 1

print("Fibonacci sequence (Iterative):")

for i in range(n):
    print(a)
    a, b = b, a + b
```

2

```
# Recursive Fibonacci Implementation

def fib(n):
    if n <= 1:
        return n
    return fib(n - 1) + fib(n - 2)


terms = int(input("Enter number of terms: "))

print("Fibonacci sequence (Recursive):")

for i in range(terms):
    print(fib(i))
```

Output

```
0
1
1
2
3
```